

A DNA Barcode Dataset for the Aquatic Fauna of the Panama Canal: Novel Resources for Detecting Faunal Change in the Neotropics

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Abstract

DNA metabarcoding is a powerful biodiversity monitoring tool, enabling simultaneous assessments of diverse biological communities. However, its accuracy depends on the reliability of reference databases that assign taxonomic identities to obtained sequences. Here we provide a DNA barcode dataset for aquatic fauna of the Panama Canal, a region that connects the Western Atlantic and Eastern Pacific oceans. This unique setting creates opportunities for trans-oceanic dispersal while acting as a modern physical dispersal barrier for some terrestrial organisms. We sequenced 852 specimens from a diverse array of taxa (e.g., fishes, zooplankton, mollusks, arthropods, reptiles, birds, and mammals) using COI, and in some cases, 12S and 16S barcodes. These data were collected for a variety of studies, many of which have sought to understand recent changes in aquatic communities in the Panama Canal. The DNA barcodes presented here are all from captured specimens, which confirms their presence in Panama and, in many cases, inside the Panama Canal. Both native and introduced taxa are included. This dataset represents a valuable resource for environmental DNA (eDNA) work in the Panama Canal region and across the Neotropics

aimed at monitoring ecosystem health, tracking non-native and potentially invasive species, and understanding the ecology and distribution of these freshwater and euryhaline taxa.

Dataset: 10.25573/data.28899749; <https://boldsystems.org>, BOLD Projects: BSFFA, BSCFA, PCIF, PCPL, PCINV, INVPA, DS-BPUSNM.

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Keywords: DNA barcode; mitochondrial DNA; fish; zooplankton; vertebrate; invertebrate; environmental DNA; biodiversity; Panama

1. Summary

Environmental DNA (eDNA) analysis using DNA metabarcoding provides novel opportunities in the biological, ecological, and geological sciences for both monitoring biodiversity and ecosystem health and looking at trophic interactions in terrestrial and aquatic environments [1]. DNA can be extracted from environmental samples without any obvious signs of biological source materials (e.g., soil, sediment, or water) as well as biological samples (e.g., fecal material, animal gut contents, or plant tissues). Entire communities can then be described through metabarcoding, which uses generic primers to amplify DNA for a given group of organisms, and next-generation sequencing techniques, making it possible to do comprehensive biodiversity surveys and monitoring programs for limited effort and costs [1–3]. Although there are constraints in the ability of this technique to detect total communities (e.g., primer mismatches or the failure to detect rare species due low DNA concentrations [4,5]), the lower cost of sample collection and the opportunity to sample more broadly have made metabarcoding an increasingly popular tool. Metabarcoding can also reduce work effort and dependency on taxonomic expertise [6], and in some cases also allows better taxonomic resolution, particularly for studies of diet description and food-webs [7].

However, taxonomically complete and reliable sequence reference databases are key to the success of metabarcoding and eDNA studies [1]. This is particularly important in species rich assemblages, such as Neotropical habitats. If the DNA reference database coverage is low or the taxonomic identifications that are available are incorrect, generated sequences get classified as unknown “Operational Taxonomic Units” (OTUs) or are misidentified, limiting the utility of the technique and the conclusions that can be drawn from metabarcoding studies. Although diversity can still be measured without knowing taxonomy assignments for sequences, accurate identifications are often needed for monitoring biodiversity effectively, making ecological inferences, or creating conservation priorities.

The Panama Canal, opened in 1914, is an artificial waterway, about 70 km in length, that connects the Eastern Pacific and Western Atlantic Oceans (Figure 1) and provides an essential conduit for global maritime trade and transport. It also creates opportunities for trans-oceanic dispersal and a modern physical dispersal barrier between eastern and western Panama for some terrestrial organisms. There are three locks on either side, which operate by gravity flow and raise and lower ships between sea level and 27 m elevation. An expansion of the Panama Canal, completed in 2016, allows larger ships to make the transit and implements water-saving basins at the new locks.

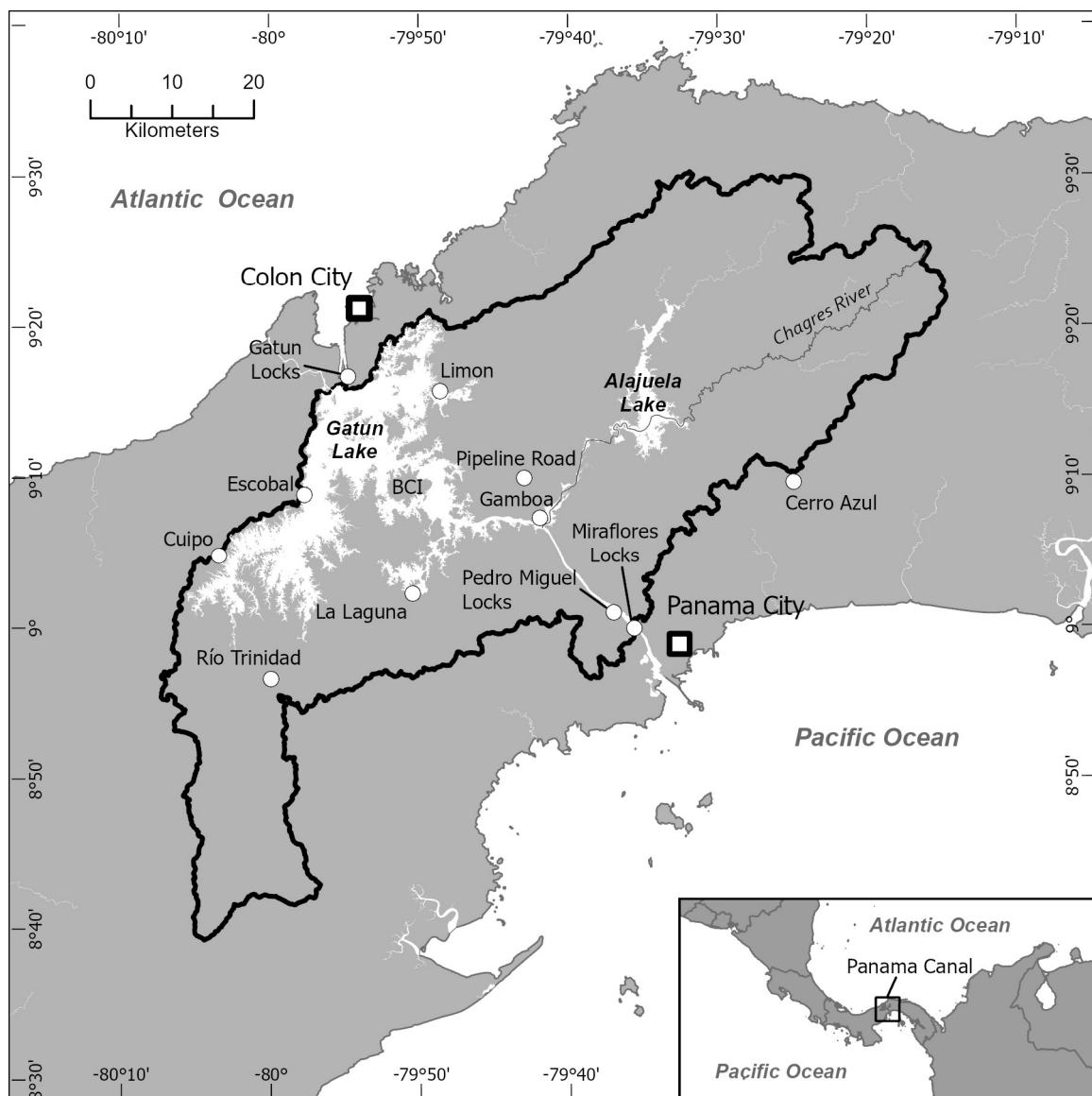


Figure 1. Sampling sites in the Panama Canal Watershed (as indicated with white circles). Names indicate locations where the majority of fish and invertebrate (e.g., plankton, arthropods, crustacea, and mollusks) specimens were collected. Gatun Locks: fishes, invertebrates; Limon: fishes; Escobal: fishes; Cuipo: fishes; Gamboa: fishes, plankton; La Laguna: fishes, invertebrates; Pedro Miguel and Miraflores Locks: fishes, invertebrates. BCI = Barro Colorado Island.

It is well understood that the communities of organisms inhabiting Gatun Lake, a large freshwater lake created when the Canal was flooded in 1913, have undergone several major shifts over time. These shifts reflect the adaptation of previously riverine species to their new lacustrine environment, the intentional introduction of non-native species, and changing management and/or abiotic conditions [8–16]. Today, ongoing research has shown that, following the recent expansion of the waterway and the opening of new locks, both physical and biological changes are taking place. Salinities appear to be increasing [17] and long-term data suggest that fish communities are changing rapidly as more predatory marine fishes are found in the Canal [3,18].

One of the most commonly used loci for animal phylogenetic and DNA barcoding analysis is the mitochondrial cytochrome c oxidase I (COI) gene [19,20]. COI has been widely sequenced as it has high interspecific variation, is easy to align, and BOLD (the Barcode of Life Database) now provides an extensive reference database [21]. However, the

cryptic morphology of many aquatic organisms makes them difficult to identify, particularly in the Panama Canal which may contain sister species from both oceans [8,11,12,18]. Further, taxonomic revisions, resulting in many species that were previously thought to have broad ranges being split into allopatric species, are now being undertaken, thus making former conspecific specimens from Mexico or Brazil not representative of a species from coastal Panama [22,23]. A recent study using eDNA to identify fishes in the Panama Canal found multiple genera that could not be resolved with COI alone. Seventeen taxa were only identified to genus and several others could not be resolved because congeners on either side of the Canal (e.g., Pacific and Atlantic Oceans), such as species in the genera *Anchoa*, *Mugil*, and *Micropogonias*, had not been previously sequenced [3].

Despite its unique geographical location and a long history of biological studies documenting the biodiversity of Panama, genetic information on many species that inhabit the Canal, or could potentially move into it, is lacking. While the COI locus database is the most complete for this region, it is still incomplete for several vertebrate and the majority of invertebrate taxa. Here we present a dataset to expand DNA barcode coverage for the aquatic fauna of the Panama Canal.

2. Data Description

The dataset consists of 792 COI, 151 12S, and 214 16S mitochondrial DNA barcodes sequenced from 852 aquatic organisms collected in Panama and, in many cases, within the Panama Canal watershed (Figure 1). Of these, 98.6% of vertebrate specimens have been identified to species level, and 68.1% of invertebrates were identified to genus and 23.1% to species, using morphological identifications confirmed by sequence similarity verification using BLAST in the NCBI database. When identification to genus or species level with high confidence was not possible (using both morphology and/or sequence variation), we report higher level taxonomy. Taxonomic coverage includes a broad range of taxa, from fishes, birds, reptiles, and mammals, to benthic and planktonic invertebrates (Table 1). Common fauna that are considered aquatic indicators in the Panama Canal, such as *Macrobrachium amazonicum* ((Heller, 1862); River prawn), *Corbicula fluminea* ((O.F. Müller, 1774); freshwater clam), *Melanooides tuberculata* ((O.F. Müller, 1774); Black snail), *Cichla monoculus* Agassiz, 1831 (Peacock bass), *Megalops atlanticus* Valenciennes, 1847 (Tarpon), and *Gambusia nicaraguensis* Günther, 1866 (Mosquitofish), are included [24]. A metadata spreadsheet with sampling information (geographic location, collector, and collection dates, etc.); GenBank Accession numbers; and BOLD project information is provided in Supplementary Table S1. Additional information regarding identification methodology, photo vouchers, and sequence chromatograms can be found for most specimens in their corresponding BOLD project records. To augment the dataset, we also include barcodes of aquatic birds and reptiles that are found in Canal habitats but were collected elsewhere in Panama. Accession numbers for barcodes of fishes that come from the surrounding oceans and have the physiological capability to enter and survive in the Canal are included in a separate worksheet in Supplementary Table S1. All sequences in fasta format and sample metadata from associated BOLD projects can also be downloaded from Figshare at: <https://figshare.com/s/7509415eab90bb93825d>.

Table 1. Taxonomic coverage of aquatic fauna in the Panama Canal Watershed and surrounding oceans for which DNA barcodes have been generated. Numbers in columns 4–6 refer to the number of barcodes available for each taxonomic group. See Supplementary Table S1 for additional taxonomic and collection details of individual samples. Photo vouchers of many specimens can be found in BOLD.

Phylum	Class	Order	COI	Locus	
				12S	16S
Arthropoda	Arachnida	Trombidiformes	3	0	0
	Branchiopoda	Anomopoda	23	0	0
		Ctenopoda	4	0	0
	Copepoda	Calanoida	2	0	0
		Cyclopoida	23	0	0
		Coleoptera	8	0	0
	Insecta	Diptera	21	0	0
		Ephemeroptera	38	0	0
		Hemiptera	54	0	0
		Lepidoptera	1	0	0
		Megaloptera	5	0	0
		Neuroptera	6	0	0
		Odonata	47	0	0
		Plecoptera	7	0	0
		Trichoptera	18	0	0
		Unknown	1	0	0
		Amphipoda	1	0	0
	Malacostraca	Decapoda	64	0	36
	Ostracoda		14	0	0
	Unknown		6	0	0
		Acanthuriformes	5	6	6
		Atheriniformes	3	2	2
		Beloniformes	6	4	4
		Blenniiformes	1	0	0
		Carangiformes	98	47	63
		Characiformes	31	14	12
		Cichliformes	18	6	7
		Clupeiformes	5	3	3
		Cyprinodontiformes	27	4	42
	Actinopterygii	Elopiformes	7	4	7
		Gobiiformes	30	11	13
		Gymnotiformes	1	0	0
		Holocentriformes	0	0	1
		Mugiliformes	4	2	2
		Perciformes	61	42	39
		Siluriformes	28	4	3
		Spariformes	1	1	1
		Synbranchiformes	3	1	1
		Syngnathiformes	2	0	1
	Chondrichthyes	Carcharhiniformes	1	0	0
		Anseriformes	3	0	0
		Charadriiformes	36	0	0
	Aves	Coraciiformes	4	0	0
		Gruiformes	2	0	0
		Pelecaniformes	5	0	0
	Mammalia	Sirenia	2	0	0
		Crocodylia	5	0	0
	Reptilia	Squamata	2	0	2
		Testudines	5	0	7

Table 1. Cont.

Phylum	Class	Order	COI	Locus 12S	16S
Cnidaria	<i>Hydrozoa</i>	Anthoathecata	4	0	0
		Mytilidae	1	0	0
Mollusca	<i>Bivalvia</i>	Unionida	2	0	0
		Venerida	2	0	0
		Architaenioglossa	3	0	0
	<i>Gastropoda</i>	Cerithiidea	1	0	0
		Lymnaeoidea	1	0	0
		Opisthorchiida	3	0	0
Platyhelminthes	<i>Trematoda</i>	Plagiorchiida	1	0	0
		Unknown	1	0	0
		Philodinida	1	0	0
Rotifera	<i>Bdelloidea</i>	Ploima	36	0	0

3. Methods

When possible, we used DNA and/or tissues from voucher specimens preserved in the collections at the Smithsonian Tropical Research Institute (STRI; Fish, Birds, and Invertebrates) and previous projects. We supplemented these collections with novel sampling in the Panama Canal, as needed. Additional fish were collected using gill nets at sampling sites throughout Gatun Lake or rotenone treatment in contained areas, such as lock chambers. Fish species that were not obtained through field sampling were purchased at local fish markets. Fin clips or muscle tissues were sampled from fish and tissues were frozen until further processing. Plankton were collected at depths of 0–3 m at two locations (near Gamboa and at or near Miraflores Lake) with several methods, including manual pumping through a 20 µm mesh or horizontal tows with a 135 µm net. Other invertebrates were collected at Miraflores and Gatun Lock chambers during maintenance events when all water was drained from the lock chambers or opportunistically during field campaigns elsewhere in Gatun Lake and Panama Canal infrastructure. All invertebrate samples were stored in 100% ethanol until DNA was extracted.

DNA was extracted from most tissue samples with the Qiagen Blood and Tissue DNA Extraction kit using the manufacturer's protocol. Plankton were extracted with a rapid lysis protocol [25]. A fragment of the 5' end of the COI gene, the approximately 670 bp Folmer region [19], was amplified for all organisms, as well as the mitochondrial 12S and 16S ribosomal RNA loci for some fish and crab species [26,27]. Primers used for each taxonomic group can be found in Table 2. PCR products were cleaned using GELase enzyme (Epicentre Biotechnologies, Madison, WI, USA) or KAPA Pure magnetic beads (Roche, Basel, Switzerland) and sequenced on an ABI 3130XL or 3500XL sequencer. Sequences were aligned and analyzed, and taxonomic identifications verified using the BLAST to NCBI option in Geneious Prime v. 2025.0.3. Lab work and sequencing was carried out in the STRI Naos Molecular Laboratory in Panama. All novel tissues (aside from plankton which were destroyed during the DNA extraction process) and extracted DNA samples have been entered into the STRI Molecular Evolution Cryo-Collection and are publicly available. Voucher specimens for many of the fishes and birds are also available in the STRI collections. Photo vouchers can be found in the various BOLD projects.

Table 2. PCR amplification and sequencing primers used for each taxonomic group.

Locus	Taxonomic Groups	Forward Primer	Reverse Primer	Source
COI	Fishes	FISH-BCL: TCAACYAAT- CAYAAAGATATYG- GCAC	FISH-BCH: ACTTCYGGGTGRCCRAARAATCA	[28]
	Birds	BirdF1: TTCTCCAACCA- CAAAGACATTG- GCAC Ltyr: TGTA AAAAGGWC- TACAGCCTAACGC	BirdR1: ACGTGGGAGATAATCCAAATCCTG COI907aH2: GTRGCNGAYGTRAARTATGCTCG	[29,30]
	Reptiles Mammals	FISH-BCL: TCAACYAAT- CAYAAAGATATYG- GCAC LCO1490: GGTCAACAAAT- CATAAAGATATTGG	FISH-BCH: ACTTCYGGGTGRCCRAARAATCA HCO2198: TAAACTTCAGGGTGACAAAAAATCA	[19,28]
	Arthropods * Crustacea Mollusks	LCO1490: GGTCAACAAAT- CATAAAGATATTGG dgLCO1490: GGTCAACAAAT- CATAAAGAYATYGG	HCO2198: TAAACTTCAGGGTGACAAAAAATCA dgHCO2198: TAAACTTCAGGGTGACCAAARAAYCA HCO2198puc: TAAACTTCWGGRTGWCCAAARAATC	[19,31,32]
	Trematodes	JB3: TTTTTTGGGCATCCT- GAGGTTTAT	COI-Rtrema: CAACAAATCATGATGCAAAAGG	[33,34]
12S	Fishes	12S229F: GYCG- GTAAAAYTCGT- GCCAG	12S954R: YCCAAGYGCACCTTCCGGTA	[27]
16S	Fishes Crustacea	16Sar: CGCCTGTTTAT- CAAAAACAT	16Sbr: CCGGTCTGAACTCAGATCACGT	[26]

* Multiple primers were mixed together to improve amplification success.

4. Scientific Relevance and Considerations

This dataset serves as a key reference that provides expanded taxonomic coverage of DNA barcodes for aquatic organisms that are found in the Panama Canal Watershed. Many of the species that are represented are closely linked in the ever-changing food web of Panama Canal's Gatun Lake and some, including mosquito fish (*Gambusia nicaraguensis*), caiman (*Caiman crocodilus fuscus* (Cope, 1868)), and river prawn (*Macrobrachium amazonicum*), are considered indicator species of freshwater habitats in the Panama Canal [24]. Several marine fishes, such as jacks (*Caranx* spp.), snooks (*Centropomus* spp.), ladyfish (*Elops* spp.), and needlefish (*Strongylura* spp.), are now increasingly common in Gatun Lake and are represented by multiple closely related species that originate from both oceans [3,18]. Threatened native taxa, such as the American crocodile (*Crocodylus acutus* Cuvier, 1807), and multiple introduced freshwater species that are now found throughout Gatun Lake, such as peacock bass (*Cichla monoculus*), Nile tilapia (*Oreochromis niloticus* (Linnaeus, 1858)), Asian clam (*Corbicula fluminea*), and trumpet snail (*Melanoides tuberculata*), are also included. In addition, we report the first observation of *Latonopsis australis* Sars, 1888 (Cladocera) in Panama. While additional sampling at more sites and across seasons will surely yield

additional taxa, particularly for plankton and other invertebrates, the dataset provides a powerful resource for future studies using DNA barcodes for species identification and long-term monitoring from environmental DNA (eDNA) and other types of biological samples, both in Panama and elsewhere in the region. In addition, advances in technology now permit the sequencing of whole mitogenomes at costs that are similar to using Sanger sequencing. We encourage future projects to generate high quality mitogenomes, rather than pursuing single locus sequencing, to provide additional options for multi-locus detections using eDNA.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/data10070108/s1>, Table S1: Sample Data.

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Data Availability Statement: All data presented in this manuscript are publicly available in BOLD and/or GenBank. GenBank Accession numbers and collection information for all sequences are provided in Supplementary Table S1. Additional information on most samples, including detailed collection information, sequences, chromatogram files, and photos of organisms, can be found in the following projects in BOLD: Fishes, BSFFA, BSCFA, PCIF, PCIFF, Invertebrates, INVPA, PCINV, PCPL. Birds, DS-BUSNM.

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Abbreviations

The following abbreviations are used in this manuscript:

STRI	Smithsonian Tropical Research Institute
ACP	Autoridad del Canal de Panamá (Panama Canal Authority)
COI	Cytochrome c oxidase subunit I
DNA	Deoxyribonucleic acid
PCR	Polymerase Chain Reaction

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